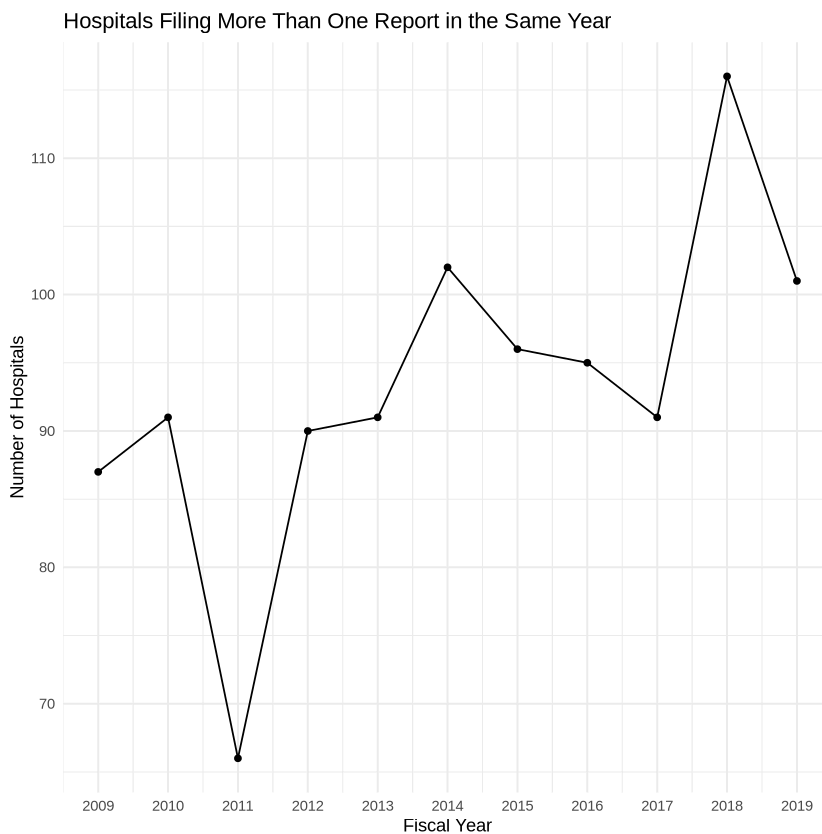


Econ 470 HW 4.1

Claire Hamilton

Question 1



Question 2

A

tibble:

1 × 1

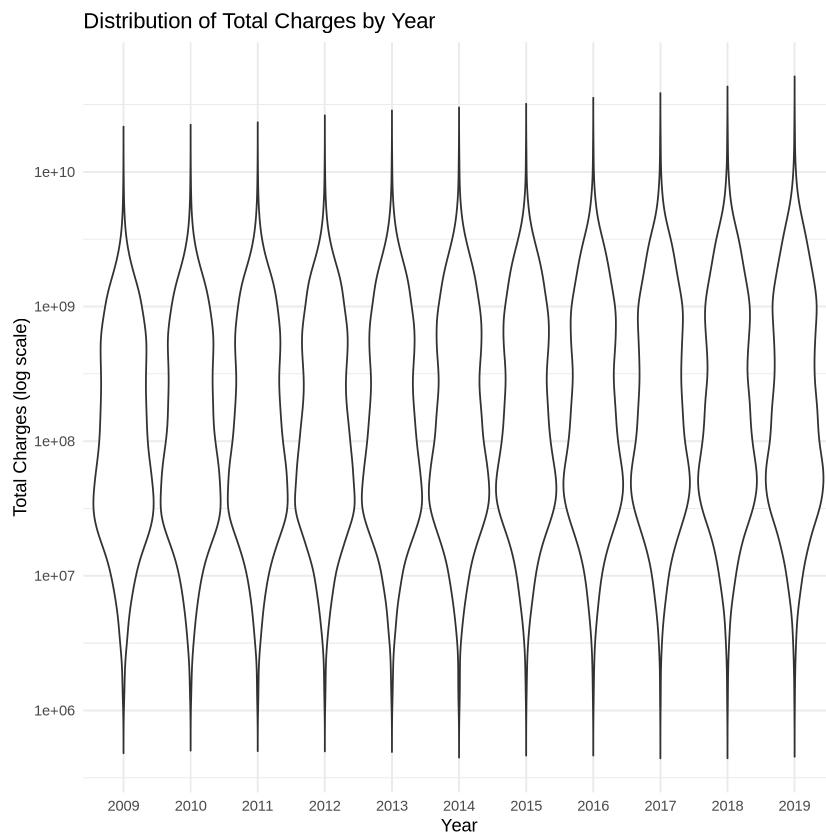
n

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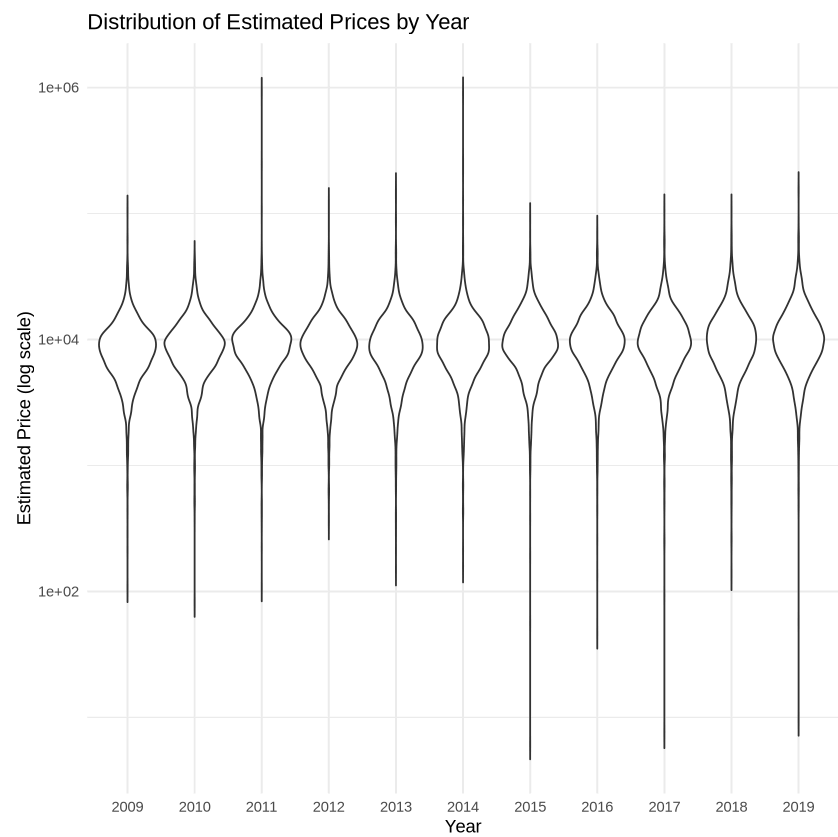
6893

After removing/combining multiple reports, 6,893 unique hospital IDs exist in the data (total, all years). I plan to create another table showing the number of unique IDs that exist in each year.

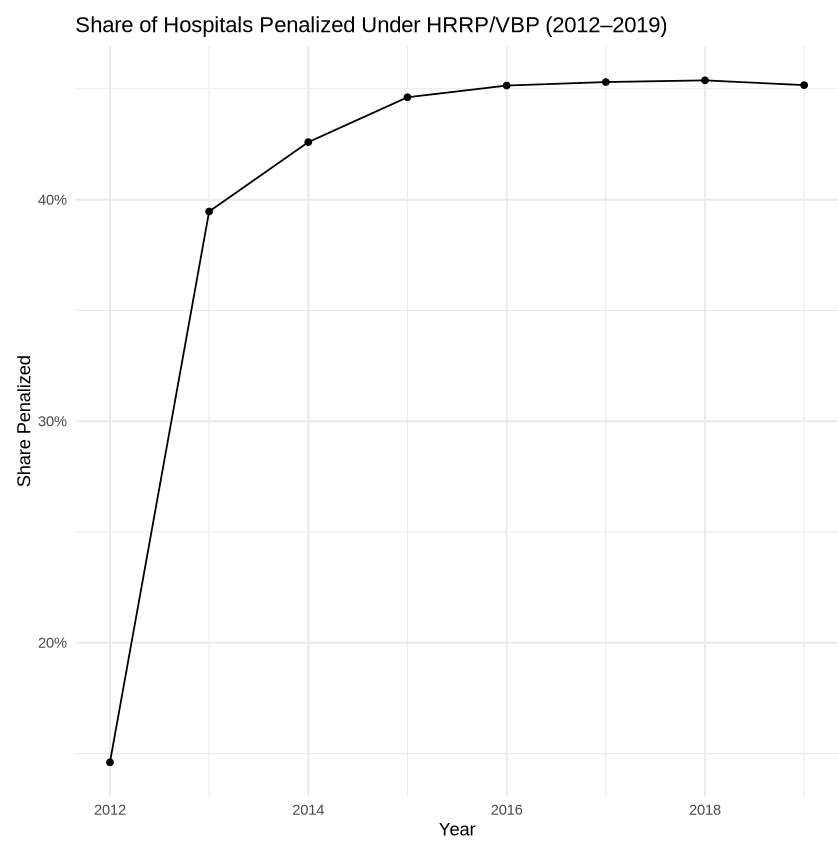
Question 3



Question 4



Question 5



Estimate ATEs

Question 6

OLS Estimates of the Effect of 2012 Net Penalties on Price Changes (2011–2014)

Dependent variable:			
	Baseline	Baseline + Bed Size	Price Change Baseline + Bed Size
	(1)	(2)	(3)
e + Medicaid			
Net Penalty (2012)	0.003 (0.009)	0.003 (0.009)	0.002 (0.010)
Mean Beds (2009–2011)		0.117 (0.239)	0.103 (0.240)
Mean Medicaid Discharges (2009–2011)			0.135 (0.175)
Constant	724.252* (380.804)	696.155* (385.199)	487.229 (471.68)
8)			
Observations	2,454	2,454	2,454
R2	0.00004	0.0001	0.0004
Adjusted R2	-0.0004	-0.001	-0.001

Note:

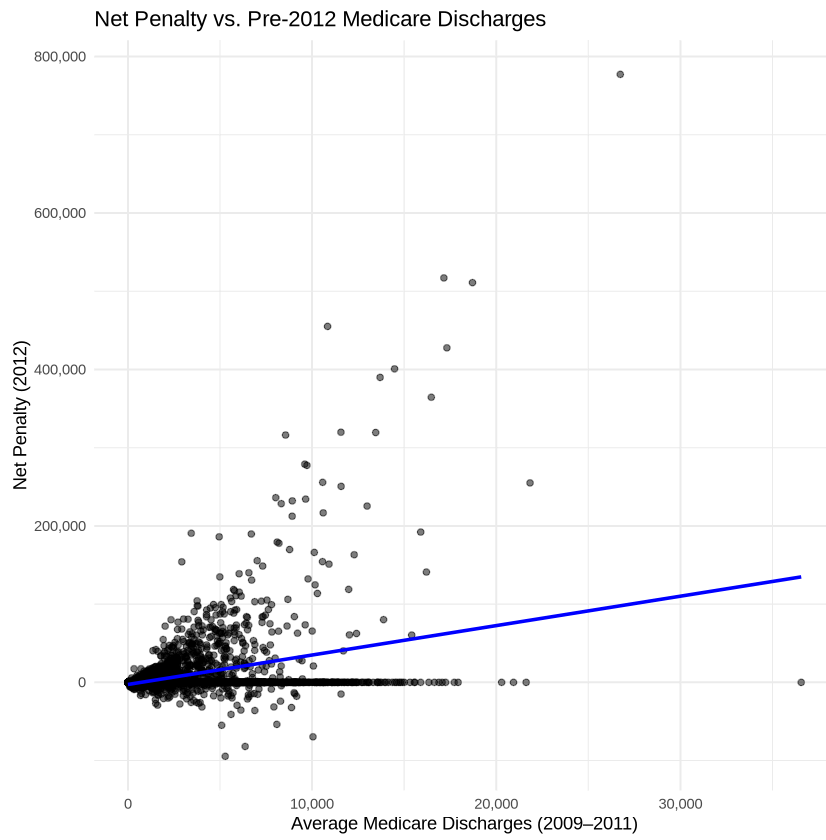
5; ***p<0.01

*p<0.1; **p<0.0

There is no statistically significant effect of net penalties on price changes for any of the models.

Question 7

```
`geom_smooth()` using formula = 'y ~ x'
```



Question 8

First Stage and Reduced Form Estimates

				Dependent variable:		
Price Change (2011–2014)				Net Penalty (2012)		
RF Baseline	RF + Beds	RF + Beds + Medicaid	FS Baseline	FS + Beds	FS + Beds + Medicaid	
(4)	(5)	(6)	(1)	(2)	(3)	
Avg Medicare Discharges (2009–2011)						
0.275**	0.271**	0.296**	4.181***	4.182***	4.591***	
			(0.237)	(0.238)	(0.273)	
(0.118)	(0.118)	(0.136)				
Mean Beds (2009–2011)						
0.061	0.064			–0.027	0.013	
				(0.483)	(0.483)	
(0.240)	(0.240)					
Mean Medicaid Discharges (2009–2011)						
–0.075					–1.227***	
					(0.401)	
(0.200)						
Constant						
–247.081	–251.124	–220.085	–5,839.040***	–5,837.246***	–5,329.239***	
			(1,138.931)	(1,139.608)	(1,149.739)	
(565.771)	(566.100)	(572.205)				
Observations						
2,454	2,454	2,454	2,454	2,454	2,454	
R2						
0.002	0.002	0.002	0.113	0.113	0.116	
Adjusted R2						
0.002	0.001	0.001	0.112	0.112	0.115	

Note:

*p<0.1; **p<0.05; ***p<0.01

The first-stage results show a statistically significant positive relationship between pre-2012 Medicare discharges and 2012 net penalties. The reduced-form estimates show that higher Medicare discharges is associated with increased price growth from 2011 to 2014.

Question 9

IV Estimates of the Effect of Net Penalties on Price Changes (2011–2014)

=====			
=====			
	Dependent variable:		

	Price Change		
	Baseline	Baseline + Beds	Baseline + Beds + Medi
caid			
	(1)	(2)	(3)

Net Penalty (2012)	0.066** (0.028)	0.065** (0.029)	0.065** (0.030)
Mean Beds (2009–2011)		0.063 (0.242)	0.063 (0.243)
Mean Medicaid Discharges (2009–2011)			0.004 (0.186)
Constant	136.455 (458.821)	127.745 (460.088)	124.003 (503.298)

Observations	2,454	2,454	2,454
R2	–0.018	–0.017	–0.017
Adjusted R2	–0.018	–0.018	–0.019
=====			
Note:	*p<0.1; **p<0.05; ***p<		
0.01			

The IV estimates indicate that higher net penalties lead to increases in hospital price growth, and the results are statistically significant. Compared to OLS, the IV estimates are smaller in magnitude. This could suggest that OLS may overstate the effect due to endogeneity that IV helps to correct for.

Question 10

The IV estimate provides a Local ATE, which shows the effect of penalties on price changes for hospitals that have penalties that are affected by their pre-2012 Medicare discharges. The LATE may differ from the overall ATE because it only applies to hospitals whose penalty status is affected by the instrument, rather than all hospitals. Thus, it may not generalize to hospitals with very low or very high Medicare discharges.